DyCo-AI

Creative Inquiry project

MISSION CONTROL

Gesture-Guided Quadruped Navigation

Fall 2026
1 credit
ME 2900
ME 3900
ME 4900

Can you guide a robot through a maze using only your hands and its onboard camera?

Your mission

Build an AI-enabled mission-control interface for the Unitree Go1 quadruped robot. Your laptop webcam will recognize hand gestures while the robot's live camera feed becomes your only view of the maze. You will transform gestures such as *forward*, *turn*, and *stop* into bounded robot velocity commands using motion planning, control design, and safety logic.

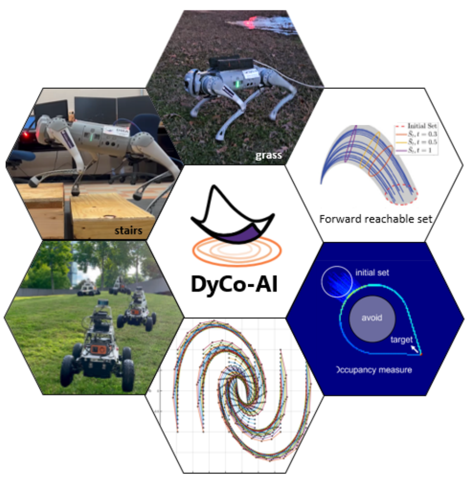
Start with webcam experiments and simulation. Finish with a live team challenge: guide the Go1 through a student-built maze safely, quickly, and without touching the walls.

Build the full robotics pipeline

PERCEIVE human intent → **Design** motion plan → **CONTROL** the Go1

The challenge is open-ended. Create new gestures, motion primitives, body poses, or gait commands—then demonstrate that your ideas remain safe and interruptible.

About the DyCo-AI Lab



The **Dynamics and Control for Autonomy and Intelligence (DyCo-AI) Lab** in Clemson Mechanical Engineering designs data-driven controllers for next-generation robots, including quadrupeds and humanoids. We combine dynamics, control, learning, and safety to help autonomous systems operate reliably in complex environments.

Target recruiting 5–6 undergraduates

Sophomores, juniors, and seniors will work primarily in teams of two. Curiosity matters more than prior coding experience; we will build the required Python and robotics skills together. Enrollment is by instructor consent.

What you will learn

- Understand how sensing, computation, planning, control, and actuation form an integrated robotic system.
- Use AI-based computer vision to recognize hand gestures and interpret human intent.
- Translate human intent into bounded robot motion using motion planning, control design, and safety constraints.
- Build, test, and communicate a human–robot teaming system through simulation and hardware experiments.

Potential benefits

- Connect core mechanical-engineering concepts to a working robotic system.
- Gain an introduction to robot perception, motion planning, control design, and human–robot teaming.
- Build skills in system integration, experimental testing, data analysis, teamwork, and technical communication.
- Present a poster at Clemson's FoCI Forum; mature work may be developed for an IEEE conference or robotics competition.

DyCo-AI Lab website

Course website

Ready to command the future?
Friday, 10:00 a.m.–12:00 noon | EIB 257 | In person

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